AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 11 · STUDENT LESSON

Training and Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Week 15: using errors to revise makes them development evidence.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Choose a split unit that matches the intended prediction claim. Try the recall before opening notes.

Your learning target

I can choose a split unit that matches the intended **prediction** claim.

I can keep related examples together and audit **group** overlap.

I can record split roles counts and limits before model evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

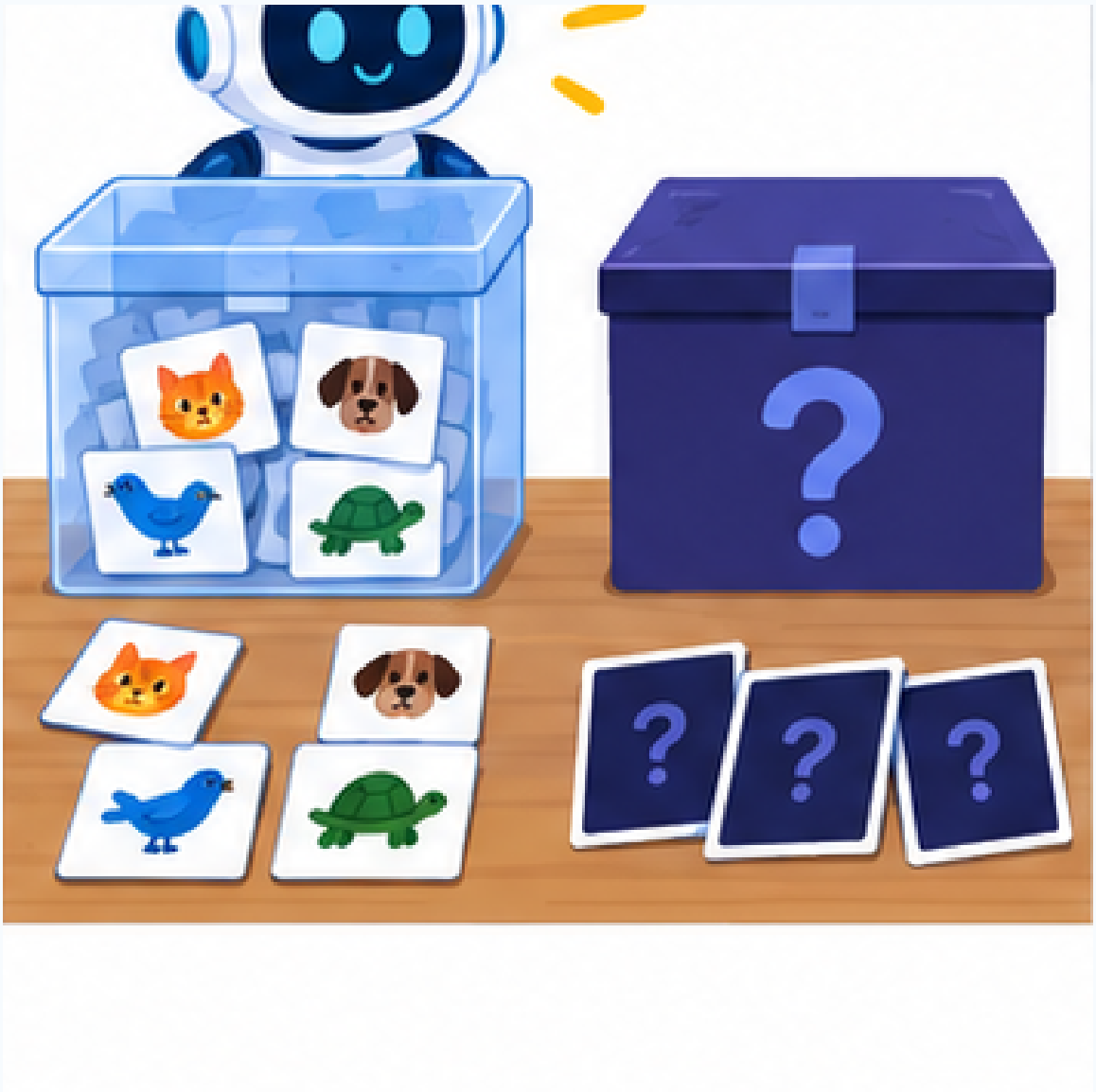
Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 11: **Training** and Testing

The **test set** estimates performance on **unseen** cases.



THINKER MISSION

Create a fair split before anyone sees the mystery cards.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Why does agreement between two annotators not prove their labels are correct?

This week's end goal

By the end, I can split related records by **group** when evaluation concerns new **group**s.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Training and Testing case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **training** and testing to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip.

They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the **group** is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions.

We therefore keep all frames from a trip in one set. To claim success on new routes, **group**ing only by trip would not be enough.

Trip group	Frame IDs	Assigned role
A	A1 A2 A3	Training
B	B1 B2	Training
C	C1	Training

Trip group	Frame IDs	Assigned role
D	D1 D2	Validation
E	E1 E2	Test
F	F1 F2	Test

This small fictional assignment is for learning split design, not **evidence** of real performance. Each frame belongs to one **group** and one set.

Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate **prediction input**.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Training and Testing. Connect each labelled input, deciding step and result.

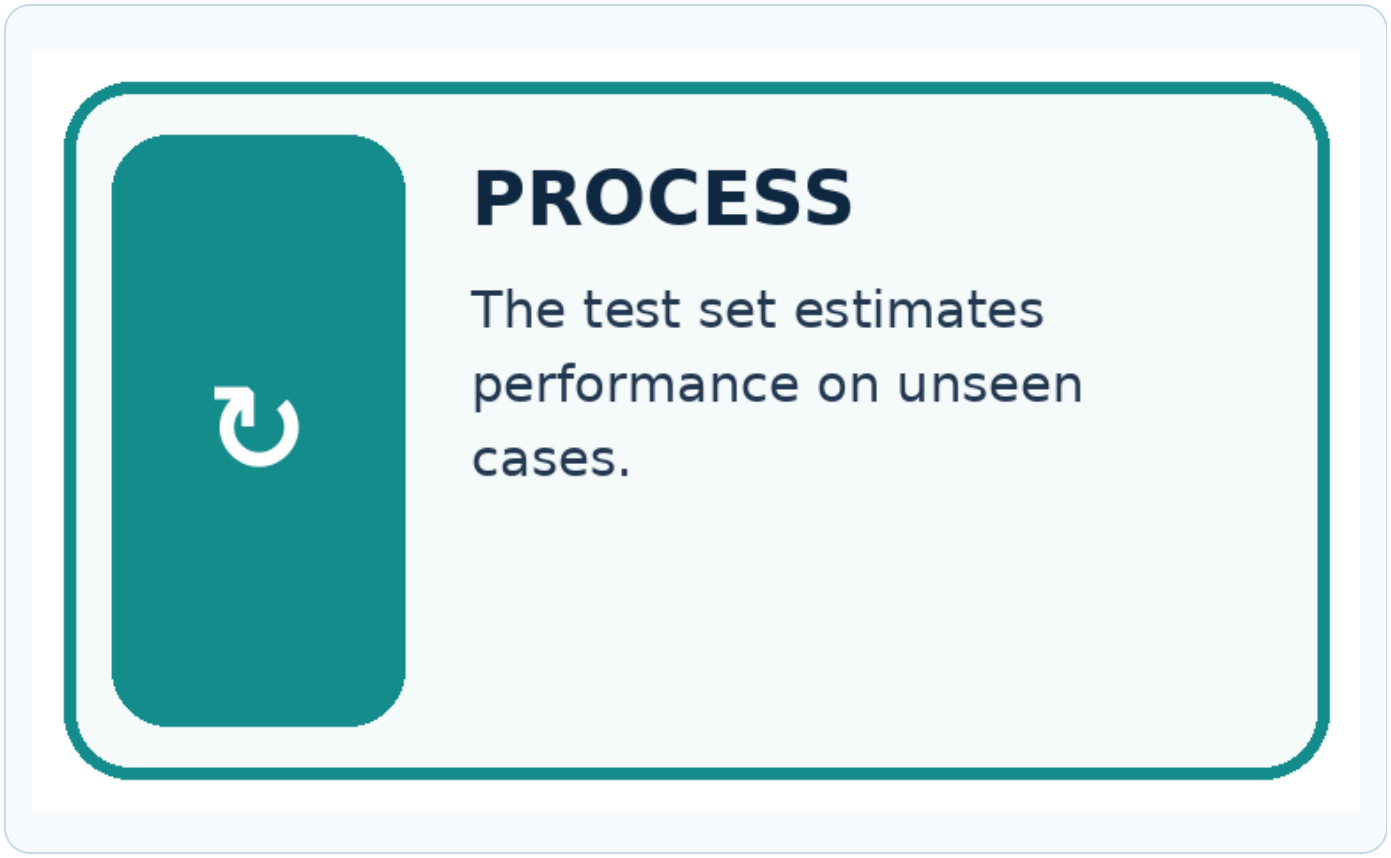
VISUAL MECHANISM

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Create a fair split before anyone sees the mystery cards.

↓


PROCESS
The test set estimates performance on unseen cases.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the **test set** for a final check after those choices are fixed.

A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

```
train_groups = {"A", "B", "C"}
valid_groups = {"D"}
```

```
test_groups = {"E", "F"}
shared = ((train_groups & valid_groups)
          | (train_groups & test_groups)
          | (valid_groups & test_groups))
print(sorted(shared))
print(len(train_groups), len(valid_groups),
      len(test_groups))
```

The first line of **output** is []: no trip ID is shared between any pair of sets. The second is 3 1 2: **training**, validation and test contain three, one and two trip **group**s. The symbol & finds shared values and | combines sets.

These are **group** counts, not frame counts.

Keep each trip in one role

WHOLE GROUPS	ROLE	PURPOSE
A B C with all frames	Training	Fit from examples
D with all frames	Validation	Compare choices
E F with all frames	Test	Final check after choices

Keep related frames together. Separate group IDs prevent this form of overlap; they do not prove that every other source of leakage is absent.

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer.

Check the assignment table and source records too.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

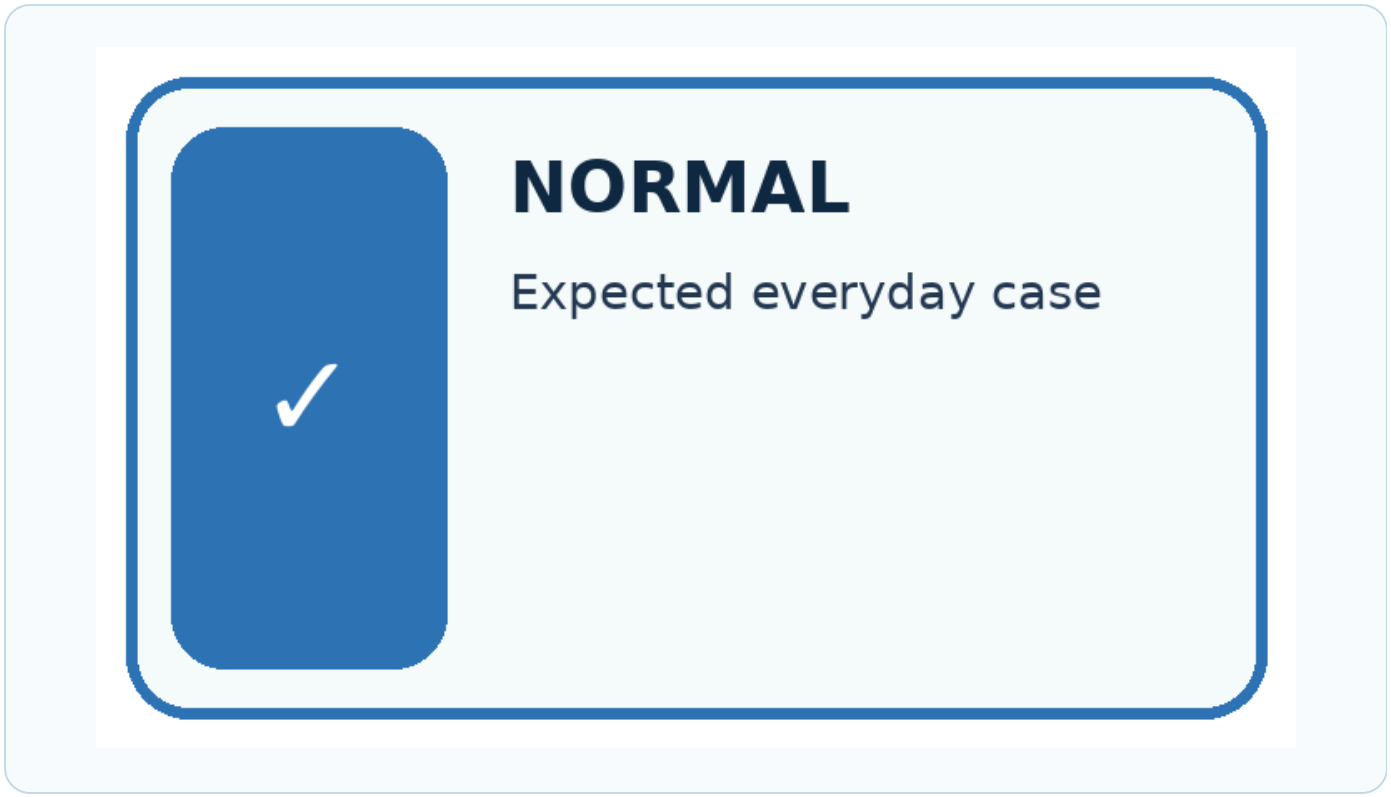
> Compare the supplied cases for Training and Testing. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

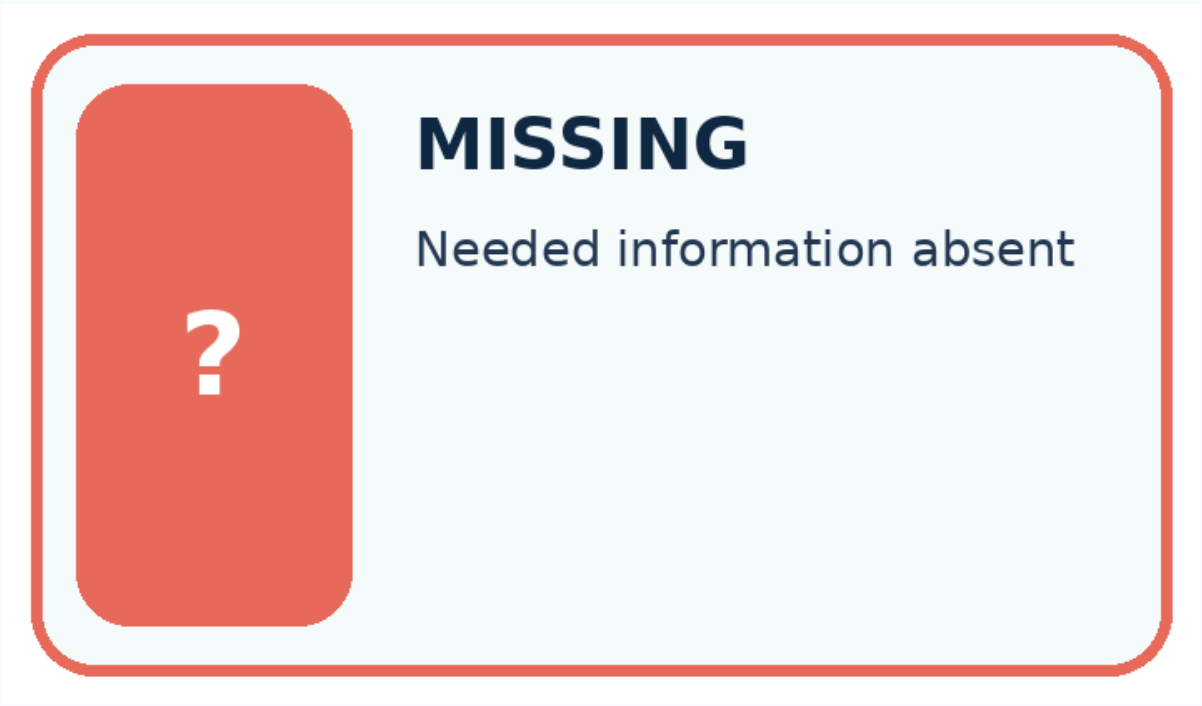
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Give A2 a new filename without changing the book it depicts.

Predicted result

Prediction: Its **group** is still A; the original related examples still cross the split.

Why the result follows

Reason: Renaming a row does not remove the relationship between scans.

STUDENT LESSON · 5 OF 18

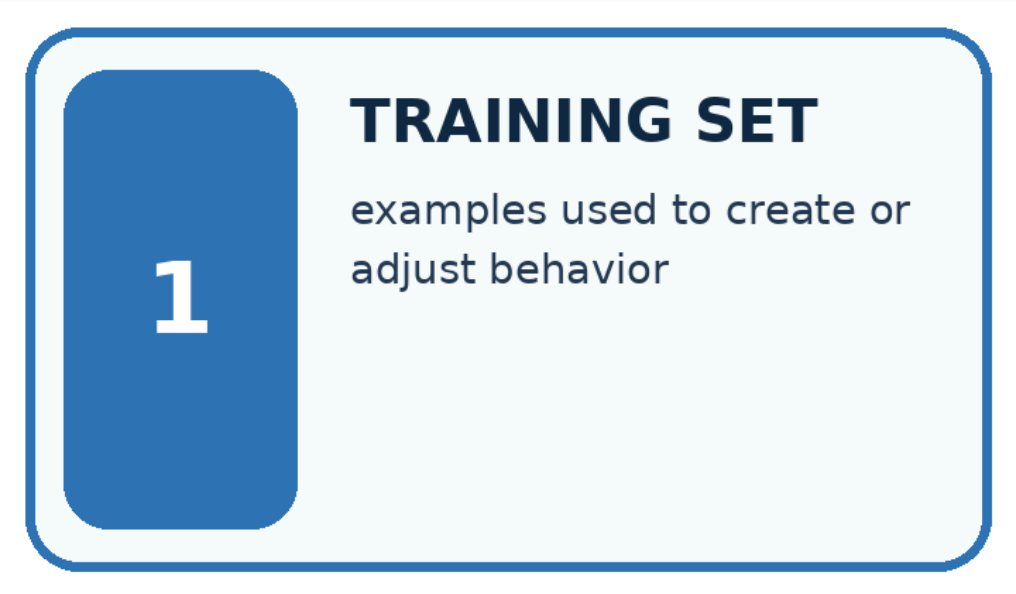
Words that unlock the idea

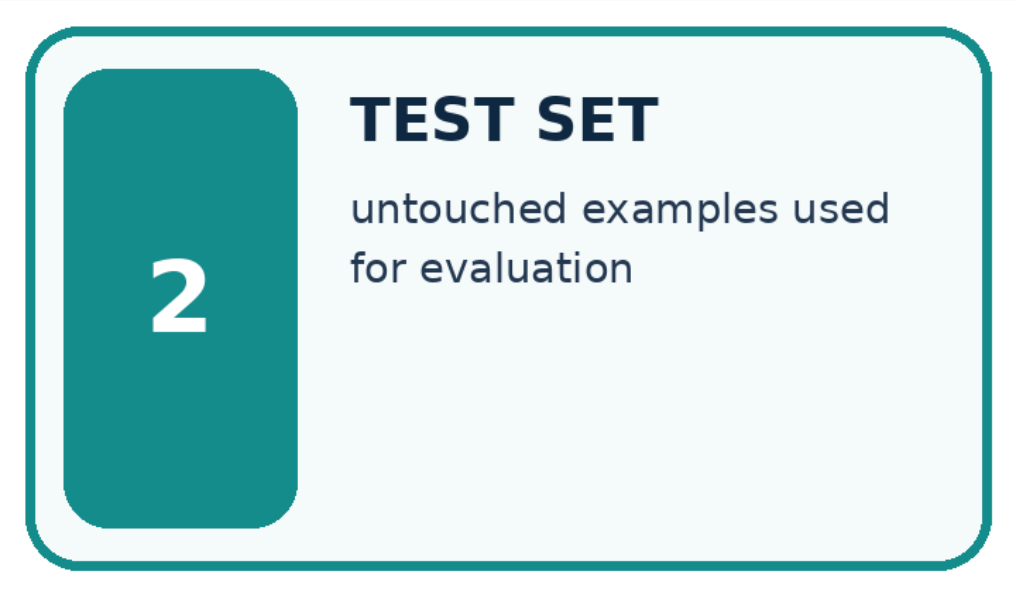
> Use the vocabulary of Training and Testing to explain a concrete decision, rather than only memorising definitions.

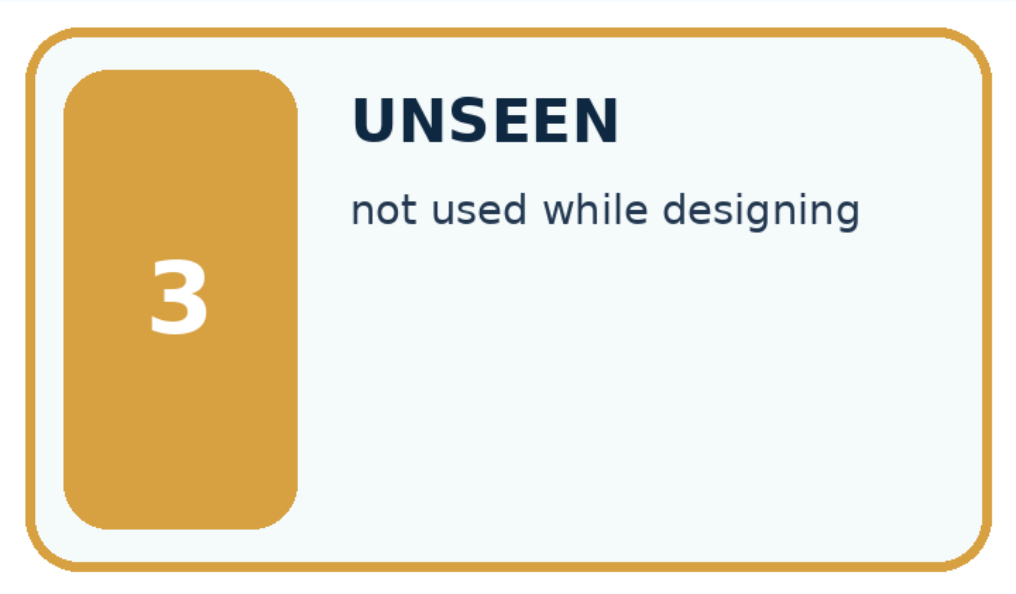
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
TRAINING SET
examples used to create or adjust behavior


2
TEST SET
untouched examples used for evaluation





Word	My own example	Not this / boundary
training set	Examples used to adjust a model's parameters or fit learned preprocessing.	
test set	Data reserved for evaluating a selected system; final test data must not influence its development.	

Word	My own example	Not this / boundary
unseen	Not previously used for the development, practice or selection being evaluated.	
leakage	Use of information during development or prediction that should be unavailable for a fair evaluation of the intended task.	

Words used in the closer explanation

Word	Meaning
Validation set	Separate examples used to compare choices during development
Group	Related examples that must stay together for a stated evaluation goal

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Training and Testing, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 11 • WORKED EVIDENCE

1. 20 cards: seal 5 as test cases first.
2. Design the rule using only the remaining 15.
3. Freeze the rule before opening the test cards.
4. If a test label guided a change, it is no longer a fair final test.

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2. Design the rule using only the remaining 15.

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4. If a test label guided a change, it is no longer a fair final test.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · Training and Testing

Suppose A1 and A2 are in **training** while A3 is in test. The frame IDs differ, but A appears in both sets. Shuffling the rows does not remove this relationship.

For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Worked explanation · Training and Testing

Audit that every source frame occurs exactly once, no **group** crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six **training**, two validation and four test: twelve total.

A fixed random seed can reproduce a random split, but it cannot make a flawed **group**ing choice fair.

A learner moves D2 from validation to training but leaves D1 in validation. Give the new row counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.

Worked explanation · Training and Testing

A split record names the **prediction** goal, source version, **group** definition, exact assignments, row and **group** counts, date and any random method or seed used.

For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 5.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training and Testing

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

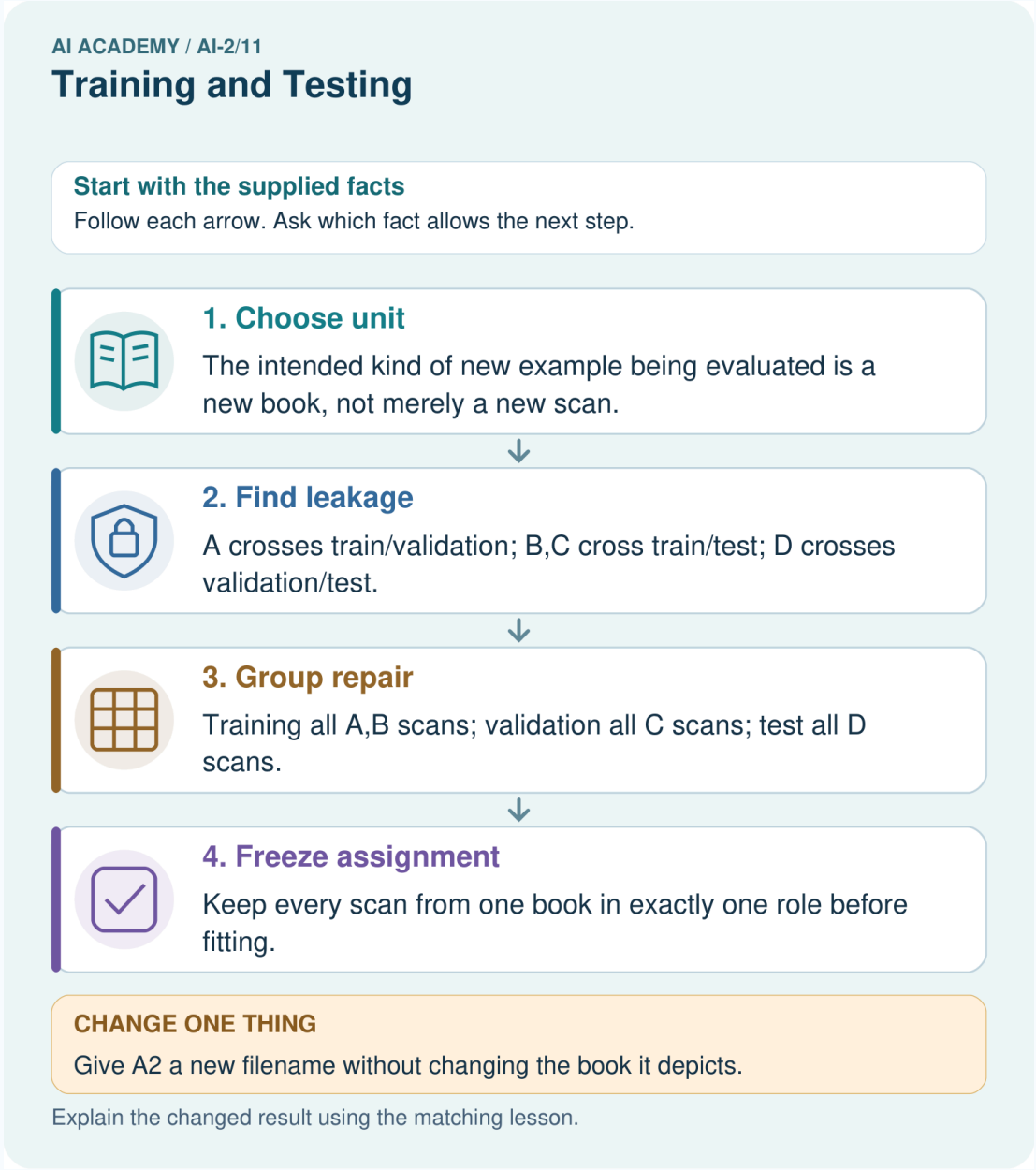
The facts we will use

Supplied facts

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Supplied facts

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Choose unit	The intended kind of new example being evaluated is a new book, not merely a new scan.
2. Find leakage	A crosses train/validation; B,C cross train/test; D crosses validation/test.
3. Group repair	Training all A,B scans; validation all C scans; test all D scans.
4. Freeze assignment	Keep every scan from one book in exactly one role before fitting.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively.

What this example does not establish

Keep the stated limit

One validation book and one test book provide very limited diversity even though no **group**s overlap.

Keep the stated limit

Read the labelled picture: How many distinct books are in each repaired role, and why report that as well as scan counts?

Keep the stated limit

Change one thing: Give A2 a new filename without changing the book it depicts. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Training and Testing claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Testing on training examples proves the system will generalize.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Fit learned pre**process**ing, such as scaling values using an average, on **training** data and apply the stored transformation to the other sets.

Using the full dataset to learn such values would let held out information influence development. We will implement pre**process**ing later; **group** separation alone does not solve that issue.

Check the claim against the facts

Record exclusions and label-review policies before scoring. Do not remove difficult test cases because they reduce the score.

If a genuine data error requires repair, document it and its effect on the evaluation rather than claiming the old test stayed untouched. Next week you will build a simple baseline using these split roles.

All trip IDs are separate, but every trip uses the same route. Can you claim the model works on unseen routes? Explain the limitation and propose a split unit for that new claim.

STUDENT LESSON · 8 OF 18

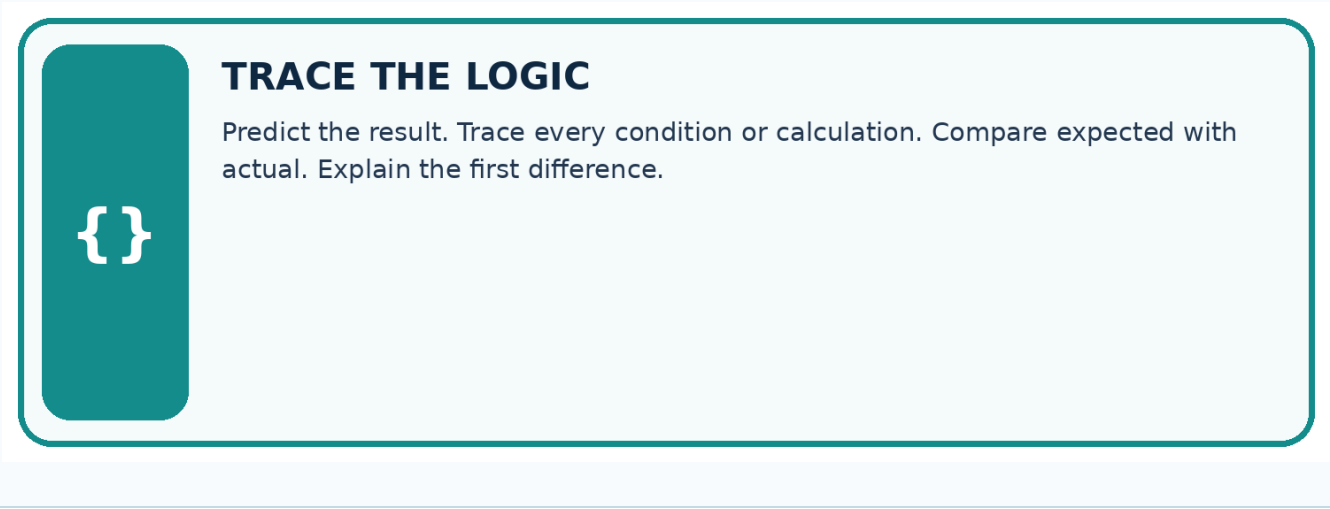
Practise together

> Practise Training and Testing with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

For new trips on a familiar route, holding out whole trips addresses within-trip similarity. For new routes, hold out entire routes so all their trips stay together.

For forecasting future events, a time-respecting split may be needed: **training** on later events to predict earlier ones would not represent that intended use. The evaluation question determines the separation.

There is no universal percentage that makes a split valid. Large **group**s can make row counts unequal.

Check that the sets can support the intended comparison, including enough relevant cases and labels, without moving individual related frames just to hit a percentage.

This twelve-row example is too small to justify a reliable performance claim.

Practise with less help: Distinguish frames from groups

Use workbook W2 for the connected example “Splits and related examples”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> How many distinct books are in each repaired role, and why report that as well as scan counts?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

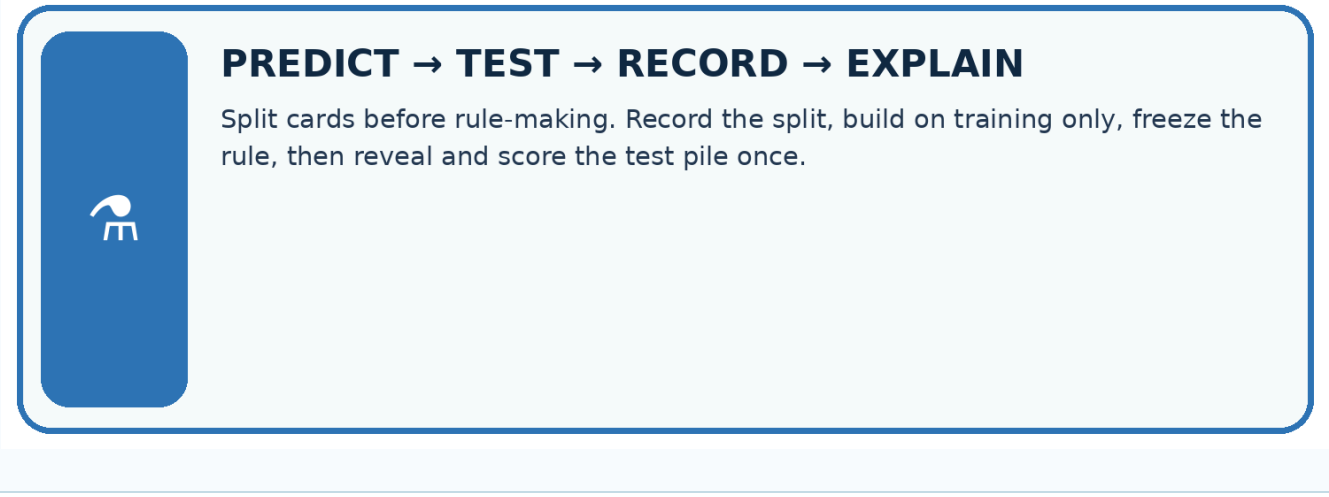
Run the investigation

> Investigate Training and Testing. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Split cards before rule-making. Record the split, build on **training** only, freeze the rule, then reveal and score the test pile once.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

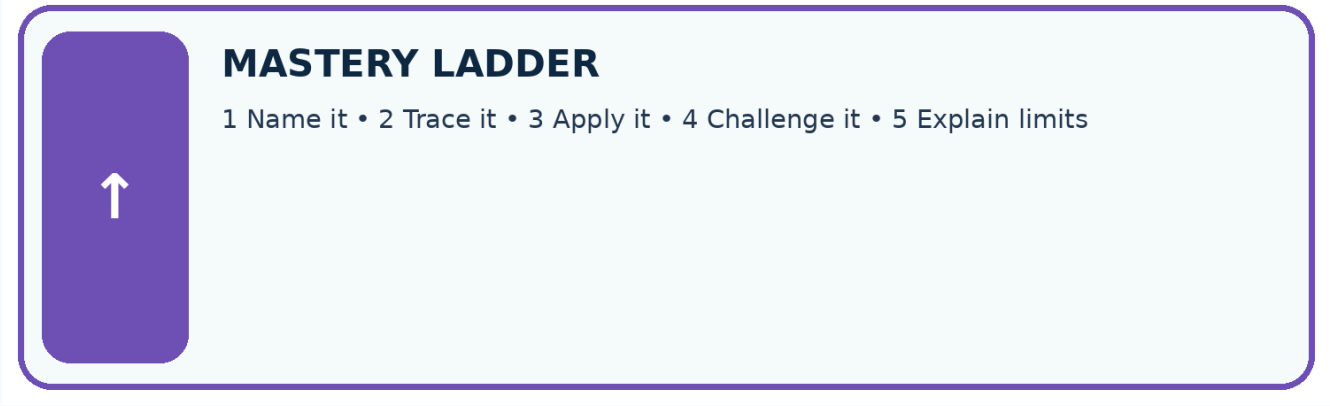
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Training and Testing. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

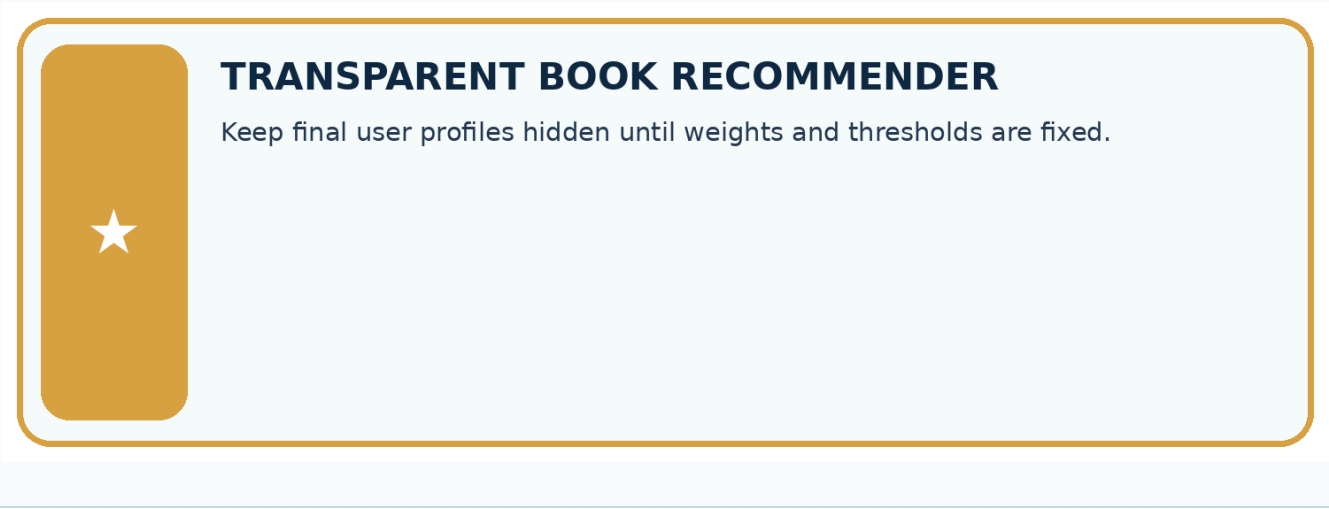
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Keep final user profiles hidden until weights and thresholds are fixed.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

STUDENT LESSON · 12 OF 18

Show what you know

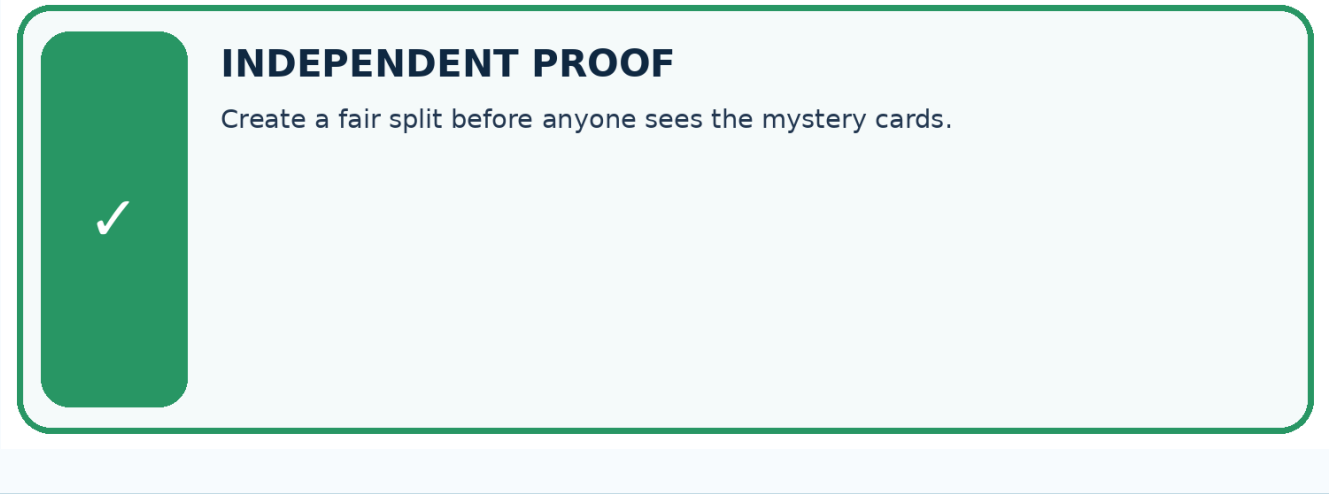
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Create a fair split before anyone sees the mystery cards.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Training and Testing case for supported practice. Repeated examples do not become fresh assessments.

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A rule is developed and evaluated using the same books A–H.

The question

> Why does this not establish **unseen** performance?

Worked explanation

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Separate development from fresh evaluation

Evidence or stage	Value or result
Books A–H	Influenced the rule
Reserved books	Did not influence construction
Revision	Keep final cases separate

A rule is developed and evaluated using the same books A–H.

> Why does this not establish **unseen** performance?

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test **evidence**?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Keep final user profiles hidden until weights and thresholds are fixed.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased **unseen** test result. Explain the issue and revise the evaluation plan.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Training and Testing, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

By the end, I can	How I will show it
Choose a split unit that matches the intended prediction claim	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Keep related examples together and audit group overlap	Explain the deciding step, show a trace or test, and state one limit.
Record split roles counts and limits before model evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Training and Testing without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Choose a split unit that matches the intended **prediction** claim

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep every scan from one book in exactly one role before **fitting**.

1. **Group** repair

2. Choose unit

3. Find **leakage**

4. Freeze assignment

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Freeze assignment Keep every scan from one book in exactly one role before **fitting**. Different scan IDs do not make examples independent books.

A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give A2 a new filename without changing the book it depicts. Which conclusion is supported by the supplied facts?

1. Its **group** is still A; the original related examples still cross the split.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its **group** is still A; the original related examples still cross the split. Renaming a row does not remove the relationship between scans.

Repair a missing building block

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

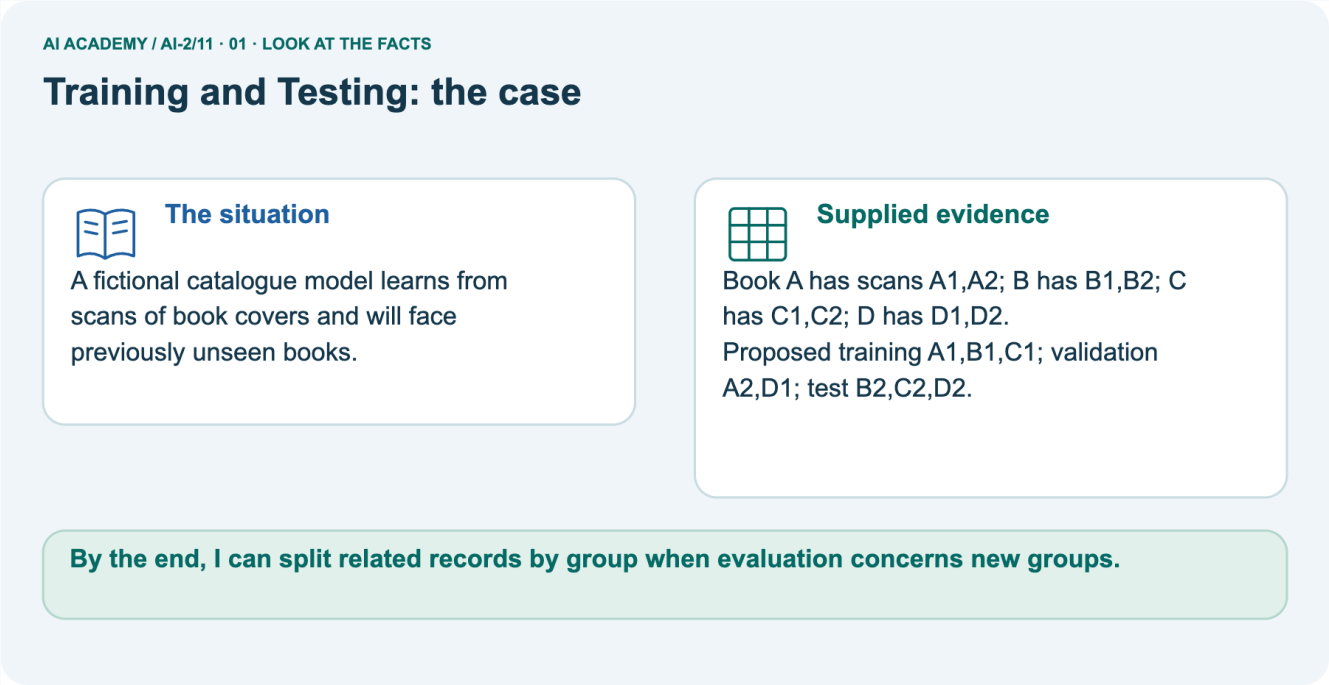
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Training and Testing graphic using its labels, deciding evidence and relevant limitation.

01. Training and Testing: the case



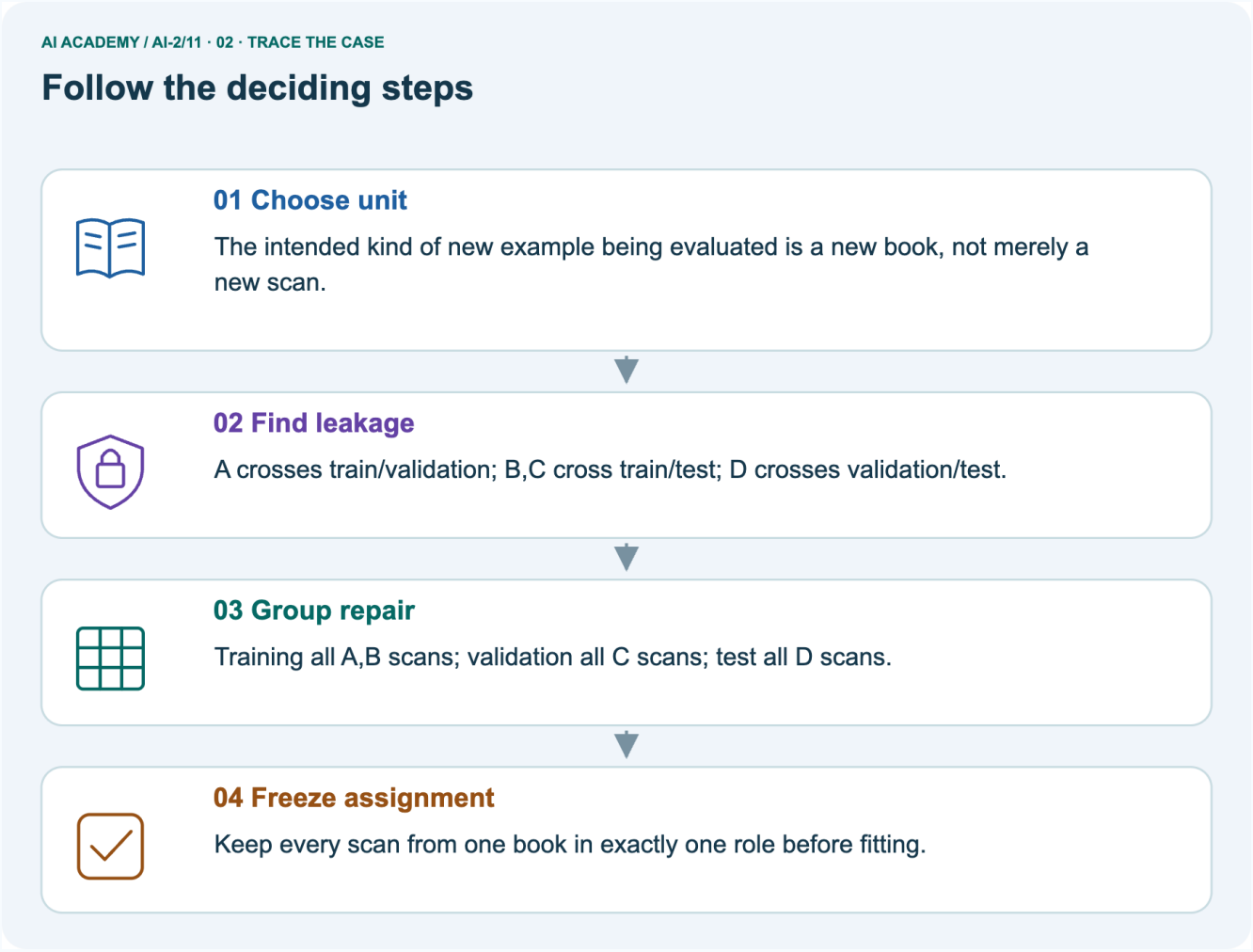
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue model learns from scans of book covers and will face previously **unseen** books. Supplied facts: Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed **training** A1,B1,C1; validation A2,D1; test B2,C2,D2.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

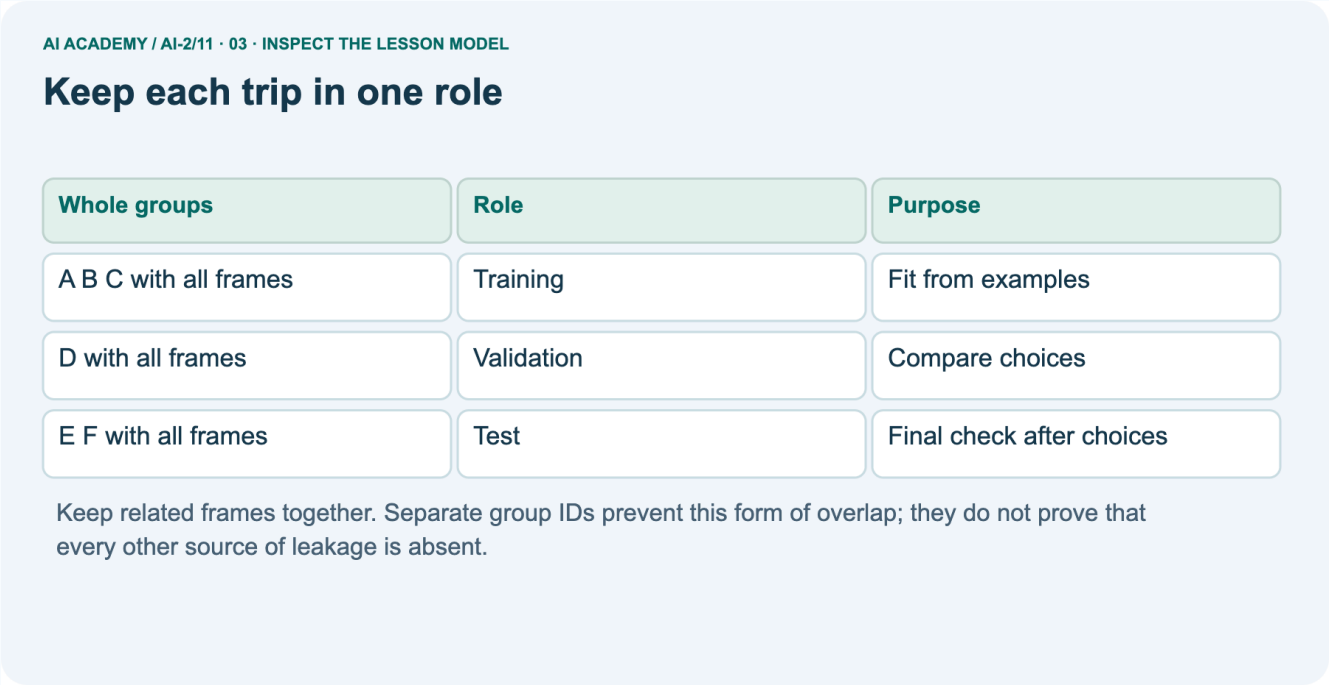


Follow the labelled picture with the matching explanation. Open the vector diagram

Choose unit: The intended kind of new example being evaluated is a new book, not merely a new scan. → Find **leakage**: A crosses train/validation; B,C cross train/test; D crosses validation/test. → **Group** repair: **Training** all A,B scans; validation all C scans; test all D scans. → Freeze assignment: Keep every scan from one book in exactly one role before **fitting**.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Keep each trip in one role

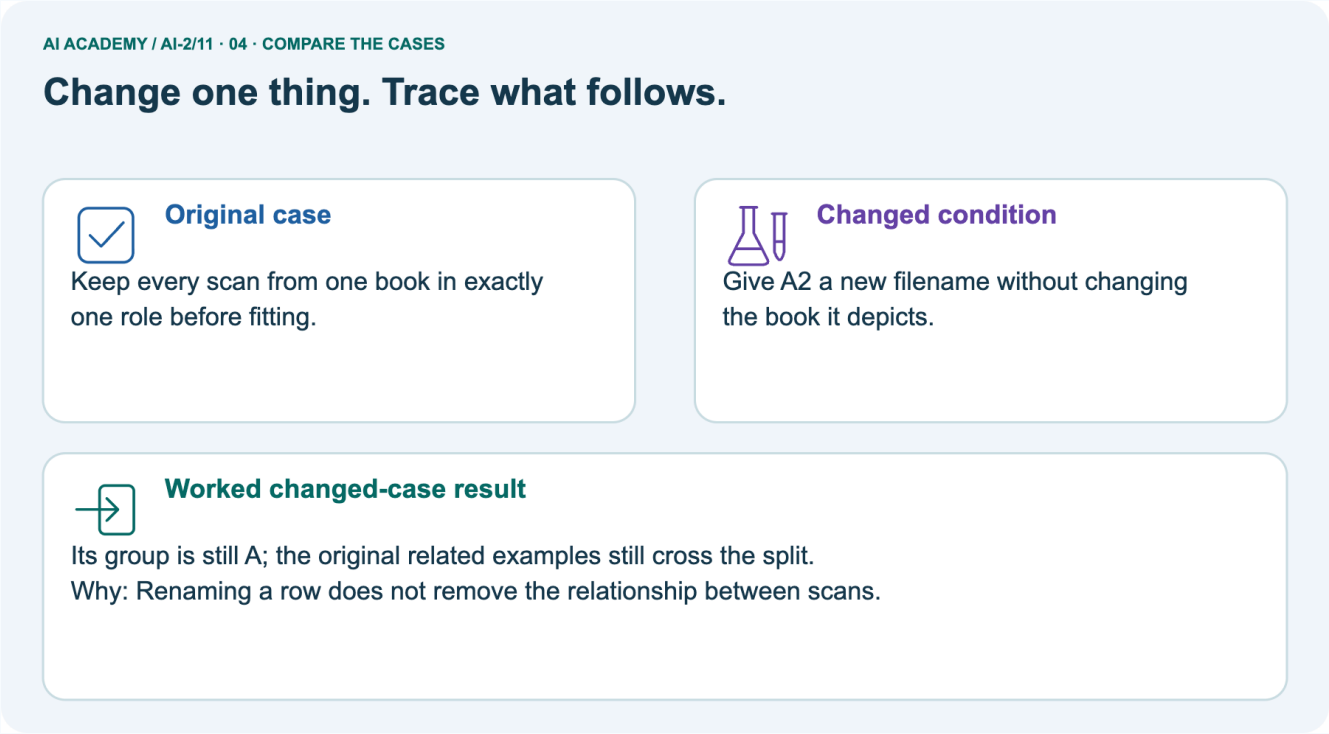


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep each trip in one role. A B C with all frames | **Training** | Fit from examples. D with all frames | Validation | Compare choices. E F with all frames | Test | Final check after choices

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



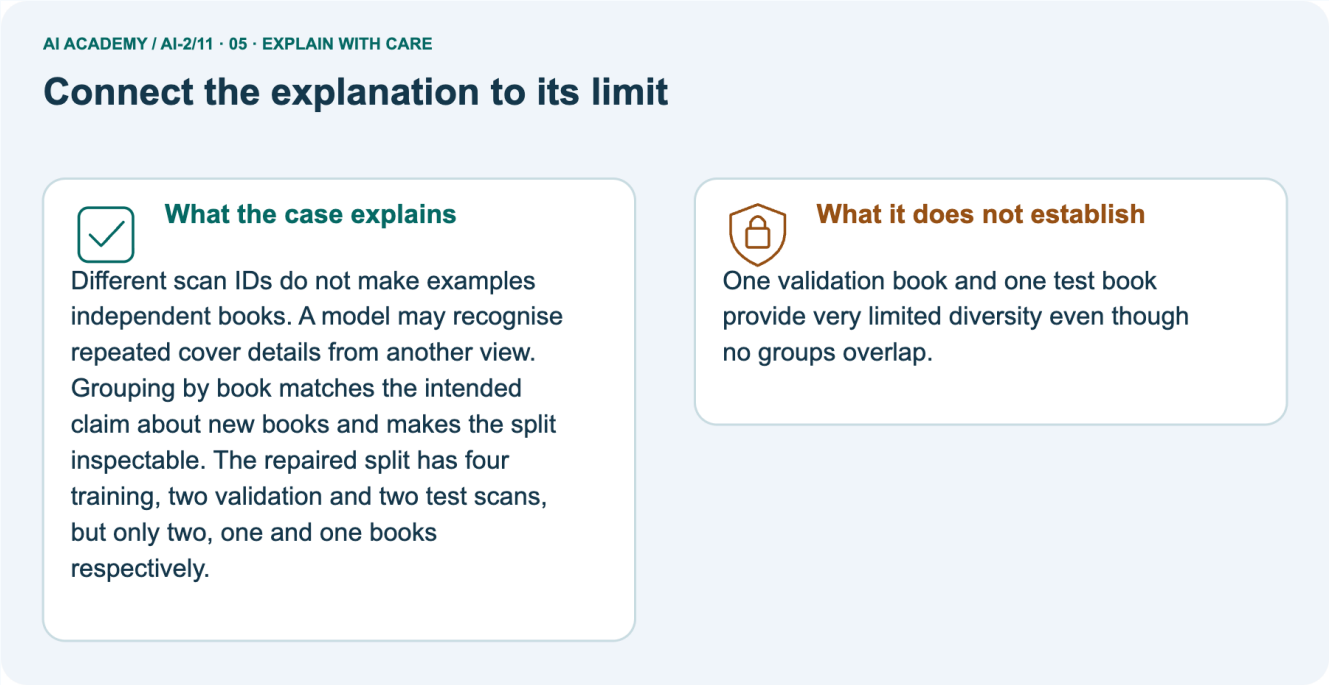
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep every scan from one book in exactly one role before **fitting**. Changed condition: Give A2 a new filename without changing the book it depicts.

Expected result: Its **group** is still A; the original related examples still cross the split. Reason: Renaming a row does not remove the relationship between scans.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



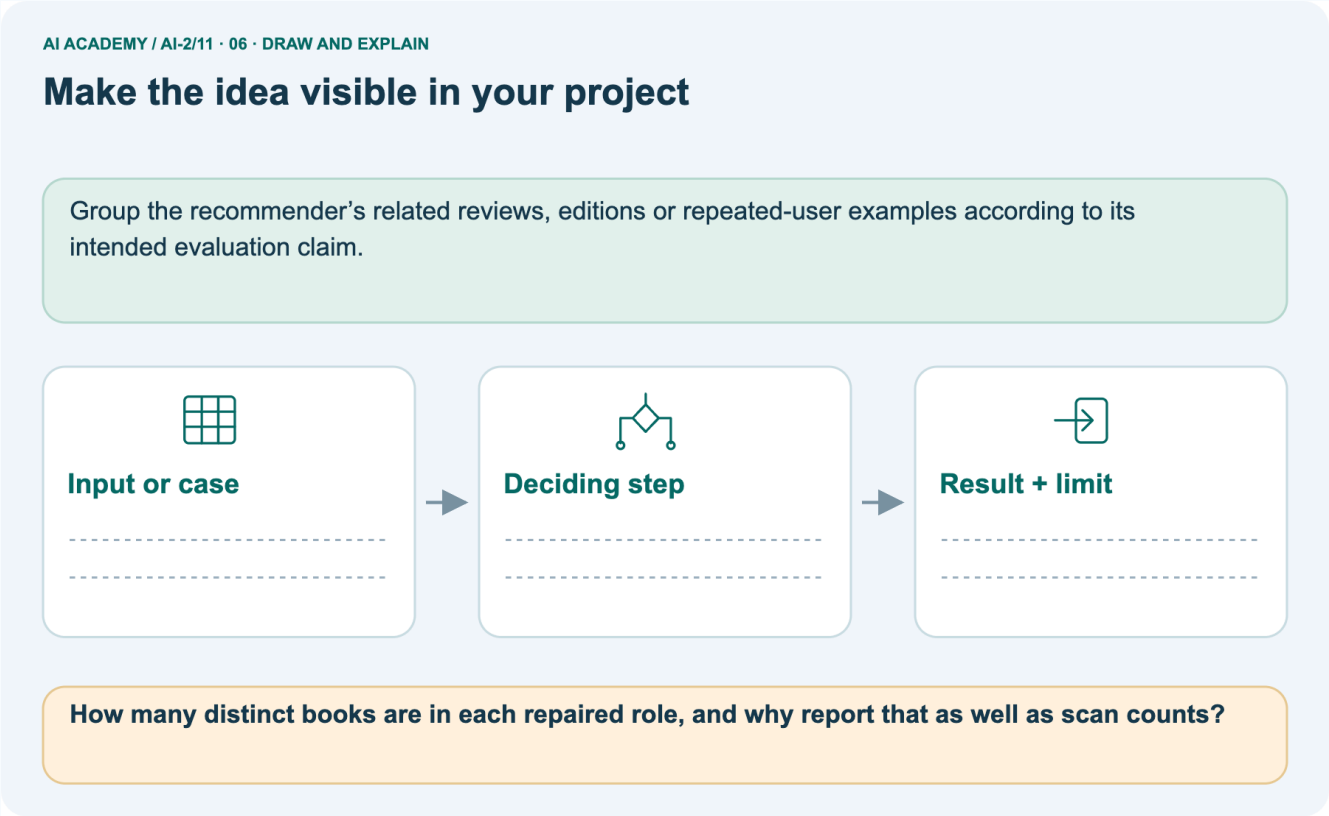
Follow the labelled picture with the matching explanation. Open the vector diagram

Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. **Group**ing by book matches the intended claim about new books and makes the split inspectable.

The repaired split has four **training**, two validation and two test scans, but only two, one and one books respectively. Limit: One validation book and one test book provide very limited diversity even though no **group**s overlap.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim. Draw your **input**, deciding step, result and limit.

Supported question: How many distinct books are in each repaired role, and why report that as well as scan counts?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.